

MODULE-1

Principles of Wireless Communication:

The Wireless Communication Environment

Communication System can be divided into two types as

- A Wireline communication System
- A Wireless communication System.

In wireline communication system, there is single-propagation path between transmitter & receiver. In this propagation medium is a coaxial cable or a twisted pair.

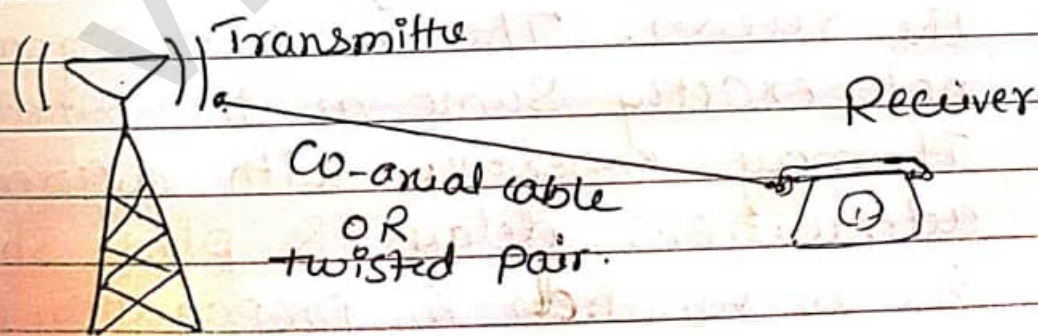


Fig: Wireline communication System.

In wireless communication system, the signal can reach the receiver via direct, reflected & scattered path as shown in below figure.

According to above figure, the ~~co~~ Superposition of multiple copies of the transmitted signal is received at the receiver. The received signal is not exactly same as transmitted signal. It may experience with different attenuations, delays & phase shifts due to variations in propagation distances & properties of scattering media.

So, in wireless receiver, there is interference of the signals received from the multiple propagation paths. & it is known as multipath interference.

The multipath interference results in an amplification or attenuation of the net received signal power observed at the receiver. This variation in the received signal strength due to multipath propagation phenomenon is known as multipath fading or simply fading.

Strong destructive interference at the receiver is referred as a deep fade & such condition may result in temporary failure of communication.

Modelling of Wireless Systems.

For better understanding, the analytical & tools & techniques are used to model & assess wireless systems.

Let us consider the transmitted passband wireless signal $s(t)$, which is transmitted across a wireless channel.

Passband signal is expressed in an analytically as follows.

$$s(t) = \text{Re} \{ s_b(t) e^{j2\pi f_c t} \}$$

Here,

$s_b(t)$ is the complex baseband representation of the transmitted signal.

To find the analytical model for the wireless communication channel, assume initially that the wireless channel is time invariant.

★ Let us consider a channel with L multipath components. Each path of the wireless channel basically has two characteristics.

(a) It delays the signal because of the propagation distance.

(b) There is an attenuation of the signal arising because of the scattering effect.

Let us consider the signal attenuation is denoted by a_i & delay of i th channel is denoted by τ_i .

Write the linear time invariant [LTI] system of impulse response is given by -

$$h_i(\tau) = a_i \delta(\tau - \tau_i)$$

The above equation is impulse response of a single path of a wireless communication system.

Wireless channel represents a linear-input-output system, the

Signal received at the receiver is the sum of the different multipath signals copies on the receive antenna.

Therefore, typical channel impulse response [CIR] of multipath-scattering based wireless channel is given by the sum of the above impulse response,

$$h(\tau) = \sum_{i=0}^{L-1} a_i \delta(\tau - \tau_i)$$

The above response is also called as the tapped delay-line model. This is called as tapped delay-line model because of the nature of the arrival of several progressively delayed components of the signal.

The above model consists of L -propagation path arising from the several reflection & scattering multipath Non-line-of-sight [NLOS] components. & One of the multipath components can also be a direct line-of-sight [LOS] component.

Each i th path is characterized by two parameters,

- (i) The attenuation factor a_i
- (ii) The path delay τ_i .

Since wireless channel is a linear time-invariant [LTI] system, the received signal $y(t)$ is convolution of transmitted signal $s(t)$ with channel impulse response [CIR] $h(t)$.

$$\therefore y(t) = s(t) * h(t)$$

$$= \int_{-\infty}^{\infty} h(\tau) s(t-\tau) d\tau$$

Substitute $\sum_{i=0}^{L-1} a_i \delta(\tau - \tau_i)$ for $h(\tau)$

$$y(t) = \int_{-\infty}^{\infty} \sum_{i=0}^{L-1} a_i \delta(\tau - \tau_i) s(t-\tau) d\tau$$

$$= \sum_{i=0}^{L-1} a_i \int_{-\infty}^{\infty} \delta(\tau - \tau_i) s(t-\tau) d\tau$$

Re-write the eqn $s(t)$.

$$s(t) = \text{Re} \{ s_b(t) e^{j2\pi f_c t} \}$$

modify eqn for $s(t-\tau)$.

$$s(t-\tau) = \text{Re} \{ s_b(t-\tau) e^{j2\pi f_c (t-\tau)} \}$$

modify the eqn for i^{th} term.

$$s(t-\tau_i) = \text{Re} \{ s_b(t-\tau_i) e^{j2\pi f_c (t-\tau_i)} \}$$

Substitute $s(t - \tau_i)$ in eqn of $y(t)$.

$$y(t) = \operatorname{Re} \left\{ \sum_{i=0}^{L-1} a_i s_b(t - \tau_i) e^{j2\pi f_c(t - \tau_i)} \right\}$$

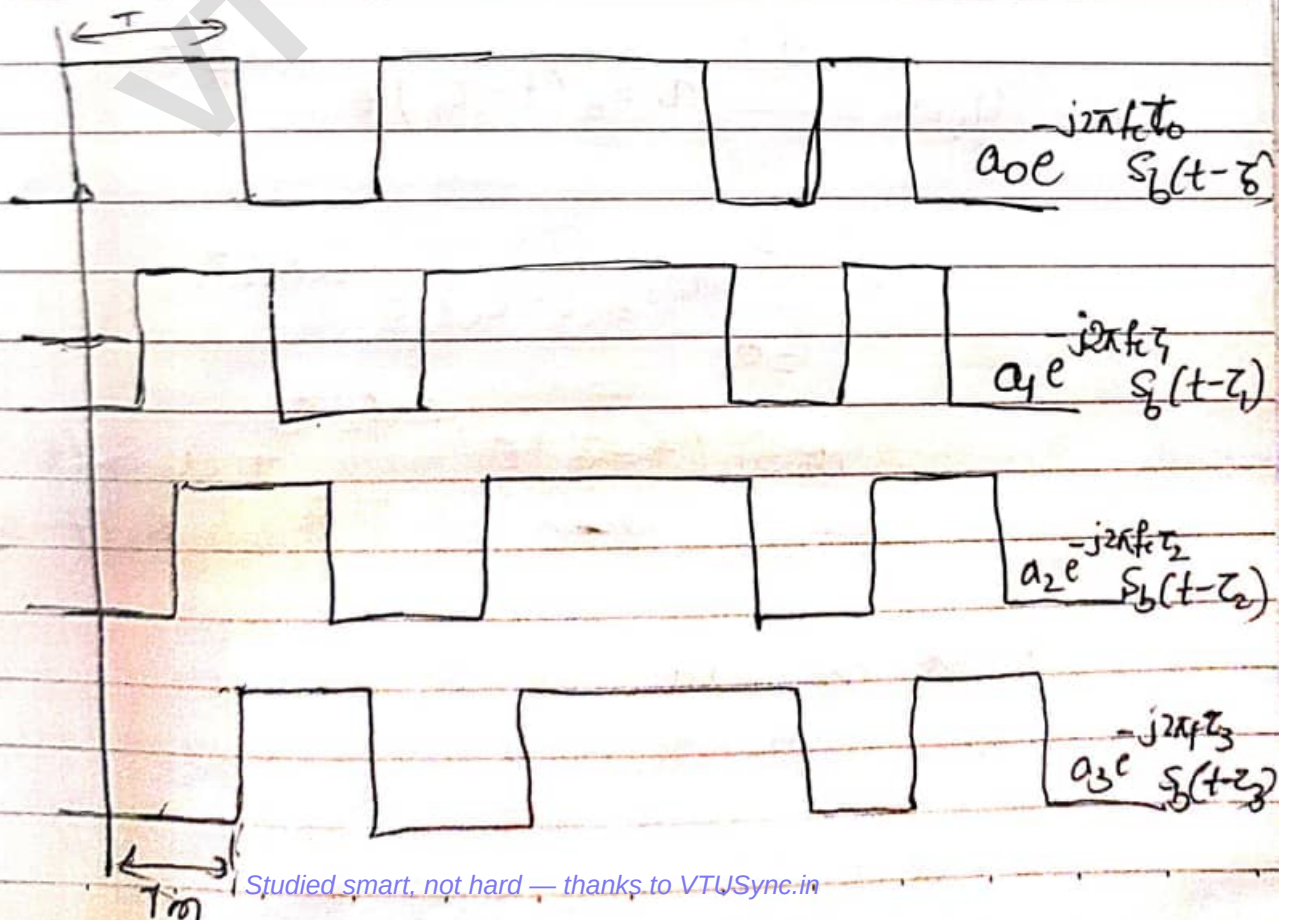
$$= \operatorname{Re} \left\{ \sum_{i=0}^{L-1} a_i s_b(t - \tau_i) e^{-j2\pi f_c \tau_i} \cdot e^{j2\pi f_c t} \right\}$$

$$= y_b(t).$$

The received signal consists of attenuation & delay parameter in passband channel.

In addition to this path delay of carrier signal corresponds to i^{th} path is given by $e^{j2\pi f_c t}$.

The different signal copies for a typical baseband BPSK information signal $s_b(t)$ is shown below.



In multipath wireless channel $T_m \ll T$

Pb: Consider a wireless signal with a carrier frequency of $f_c = 850 \text{ MHz}$, which is transmitted over a wireless channel that results in $L=4$ multipath components at delays of 201, 513, 819, 1223 ns & corresponding to received signal amplitudes of 1, 0.6, 0.3, 0.2 respectively. Derive the expression for the received baseband signal $y_b(t)$ if the transmitted baseband signal is $s_b(t)$.

Write the expression of $y_b(t)$.

$$y_b(t) = \sum_{i=0}^{L-1} a_i s_b(t - \tau_i) e^{-j2\pi f_c \tau_i}$$

Substitute $L=4$

$$y_b(t) = \sum_{i=0}^{4-1} a_i s_b(t - \tau_i) e^{-j2\pi f_c \tau_i}$$

$$= \sum_{i=0}^3 a_i s_b(t - \tau_i) e^{-j2\pi f_c \tau_i}$$

$$= a_0 s_b(t - \tau_0) e^{-j2\pi f_c \tau_0} + a_1 s_b(t - \tau_1) e^{-j2\pi f_c \tau_1} \\ + a_2 s_b(t - \tau_2) e^{-j2\pi f_c \tau_2} + a_3 s_b(t - \tau_3) e^{-j2\pi f_c \tau_3}$$

Find the value of the following

$$a_0 e^{-j2\pi f_c \tau_0}$$

$$a_1 e^{-j2\pi f_c \tau_1}$$

$$a_2 e^{-j2\pi f_c \tau_2}$$

$$a_3 e^{-j2\pi f_c \tau_3}$$

$$a_0 e^{-j2\pi f_c \tau_0} = 1 \times e^{-j2\pi 850 \times 10^6 \times 201 \times 10^{-9}}$$

$$= 0.59 + j0.81$$

$$\text{U4 } a_1 e^{-j2\pi f_c \tau_1} = 0.6 \times e^{-j2\pi 850 \times 10^6 \times 513 \times 10^{-9}}$$

$$= 0.18 - j0.24 \quad 0.57 - j0.19$$

$$a_2 e^{-j2\pi f_c \tau_2} = 0.3 \times e^{-j2\pi 850 \times 10^6 \times 819 \times 10^{-9}}$$

$$= 0.18 - j0.24$$

$$a_3 e^{-j2\pi f_c \tau_3} = 0.2 \times e^{-j2\pi 850 \times 10^6 \times 1223 \times 10^{-9}}$$

$$= -0.19 + j0.06$$

$$y_b(t) = (0.59 + j0.81) S_b(t - 201 \times 10^{-9}) + (0.57 - j0.19) S_b(t - 513 \times 10^{-9})$$

$$+ (0.18 - j0.24) S_b(t - 819 \text{ ns}) + (-0.19 + j0.06) S_b(t - 1223 \times 10^{-9})$$

System model for Narrowband Signal

In narrowband signal, different delayed components $S_b(t - \tau_i)$ are approximately equal to each other.

$$\text{i.e. } S_b(t - \tau_i) \approx S_b(t)$$

Such wireless channel is also known as a flat-fading wireless channel.

∴ for a narrowband transmit signal $s_b(t)$ the expression for the received baseband signal $y_b(t)$ is given by.

$$y_b(t) = \left[\sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i} \right] S_b(t)$$

$$= h S_b(t) \quad a e^{j\phi}$$

where $h = a e^{j\phi}$

$$= \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i}$$

is termed as the complex fading channel coefficient.

$\therefore$ O/P baseband signal is related to the i/p baseband signal $S_b(t)$ by complex factor $e^{j\phi}$.

For instant, consider two complex component multipath channel with identical magnitude & exactly out of phase components. $a_0 = a_1$ & $e^{-j2\pi f_c \tau_0} = -e^{-j2\pi f_c \tau_1}$

In this case the received signal

$$y_b(t) = \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i}$$

In narrowband
 $\tau_0 \approx \tau_1 \approx \tau$

$$= a_0 e^{-j2\pi f_c \tau_0} + a_1 e^{-j2\pi f_c \tau_1}$$

$$= a_0 e^{-j2\pi f_c \tau} - a_1 e^{-j2\pi f_c \tau}$$

$$= a_0 e^{-j2\pi f_c \tau} - a_0 e^{-j2\pi f_c \tau}$$

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So narrowband approximation is not used ~~in~~ hold in a wideband system such as a CDMA-based one.

for problem ①, derive the corresponding received signal with the narrowband assumption.

$$\begin{aligned}y(t) &= \left[\sum_{i=0}^3 a_i e^{-j2\pi f_c \tau_i} \right] S_b(t) \\&= \left[a_0 e^{-j2\pi f_c \tau_0} + a_1 e^{-j2\pi f_c \tau_1} + a_2 e^{-j2\pi f_c \tau_2} + a_3 e^{-j2\pi f_c \tau_3} \right] S_b(t) \\&= \left[(0.59 + j0.81) + (0.57 - j0.19) + (0.18 - j0.24) + (-0.19 + j0.06) \right] S_b(t) \\&= [1.15 + j0.44] S_b(t).\end{aligned}$$

So, Net received baseband signal is expressed as

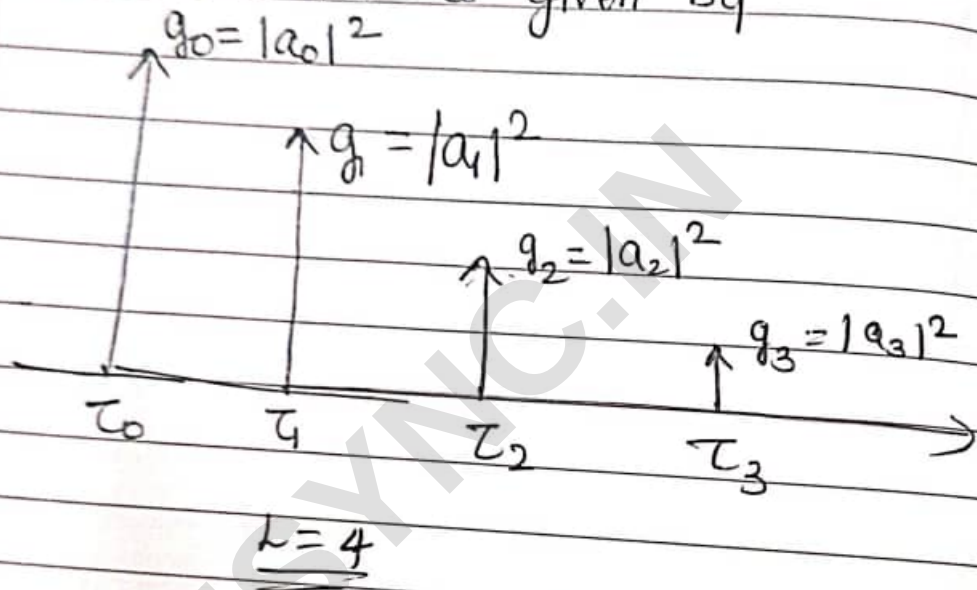
$$y_b(t) = h S_b(t).$$

h = Channel coefficient $h = 1.15 + j0.44$

Basic wireless channel modeling

The wireless channel consists of many multipath components due to the NLOS signal propagation path.

Inter-symbol interference & time-varying characteristics is given by



Write the expression for impulse response of standard multipath wireless channel.

$$h(t) = \sum_{i=0}^{L-1} a_i \delta(t - \tau_i)$$

$\delta(t - \tau_i)$ delaying the signal by τ_i
 a_i is attenuation associated with i th path.

In general power response of impulse response is given by

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$$P(\tau) = |h(\tau)|^2$$

where $h(\tau)$ is the impulse response at delay τ .

consider $\phi(t)$ is used to represent multipath power profile of the multipath channel.

$$\phi(t) = \sum_{i=0}^{L-1} |a_i|$$

$$\phi(t) = [h(t)]^2$$

$$= \left[\sum_{i=0}^{L-1} a_i \delta(t-\tau_i) \right]^2$$

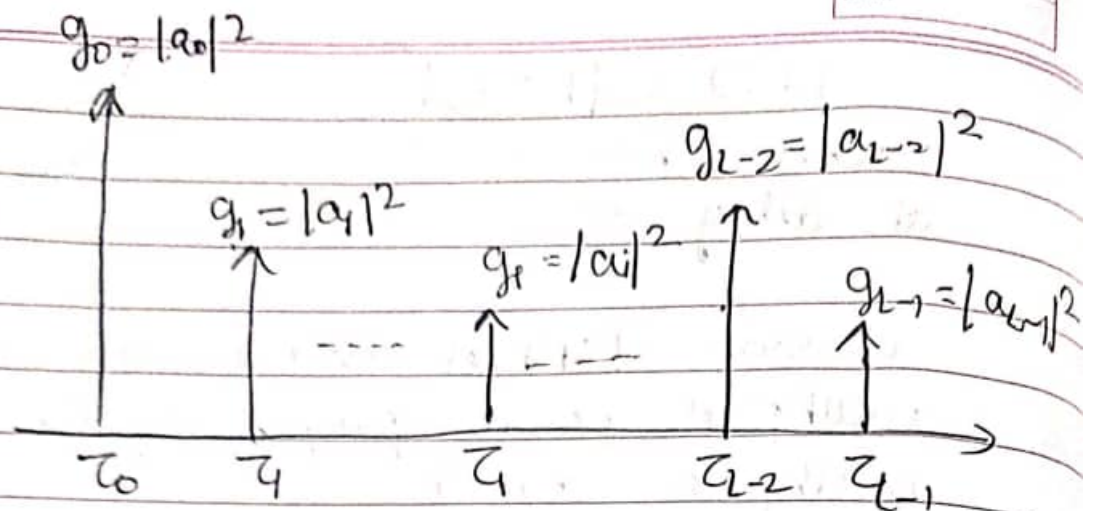
$$= \sum_{i=0}^{L-1} |a_i|^2 \delta(t-\tau_i)$$

$$= \sum_{i=0}^{L-1} g_i \delta(t-\tau_i)$$

Here $g_i = |a_i|^2$ is the power gain of the i^{th} path.

The total power received in a multipath wireless channel occurs over a spread of time referred to as the delay spread.

The spread of the arriving power at the wireless receiver is schematically shown as below.



maximum Delay Spread $\sigma_T^{\max}$

The framework to quantify the delay spread of a wireless channel is through the maximum delay spread of the channel & is denoted by $\sigma_T^{\max}$.

$$\sigma_T^{\max} = \tau_{L-1} - \tau_0$$

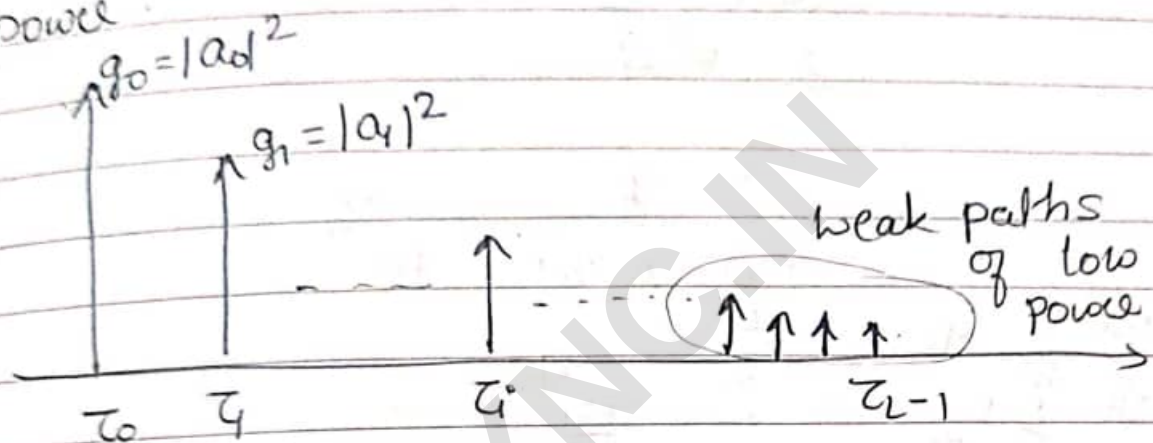
$\uparrow$ $\uparrow$
 Last signal at delay of τ_0 first path arriving
 copying arriving
 at τ_{L-1}

Rms delay spread σ_T^{rms}

In wireless channel, the paths which arrive late are significantly lower in power due to large propagation distance & weaker reflection.

In this case maximum delay spread is not reliable to indicate the true power spread of the arriving multipath signal components.

So, Rms delay spread is more realistic indicator for the spread of the signal power.



Consider the power profile comprising of L multipath components with $g_i = |a_i|^2$

$0 \leq i \leq L-1$ denoting the power gain of each multipath component.

Consider a new quantity b_i as

$$b_i = \frac{g_i}{g_0 + g_1 + \dots + g_{L-1}}$$

where g_i is total power corresponds to i^{th} path

$g_0 + g_1 + g_2 + \dots + g_{L-1}$ denotes the total power in the multipath power profile.

The ratio b_i denotes the fraction of power in the i th multipath component.

$\therefore$ Average delay $\bar{\tau}$ can be determined by

$$\bar{\tau} = b_0 \tau_0 + b_1 \tau_1 + \dots + b_{L-1} \tau_{L-1}$$

$$= \sum_{i=0}^{L-1} b_i \tau_i$$

$$= \sum_{i=0}^{L-1} \frac{g_i}{(g_0 + g_1 + \dots + g_i)} \tau_i$$

change
as $\sum_{j=0}^{L-1} g_j$

$$= \sum_{i=0}^{L-1} \frac{g_i}{\sum_{j=0}^{L-1} g_j} \tau_i$$

Average delay $\bar{\tau}$ is obtained by weighting each delay τ_i in proportion to the fraction of the power b_i .

Rms delay is computed by standard deviation of the power b_i .

$$(\sigma_{\tau}^{\text{rms}})^2 = b_0 (\tau_0 - \bar{\tau})^2 + b_1 (\tau_1 - \bar{\tau})^2 + \dots + b_{L-1} (\tau_{L-1} - \bar{\tau})^2$$

$$= \sum_{i=0}^{L-1} b_i (\tau_i - \bar{\tau})^2$$

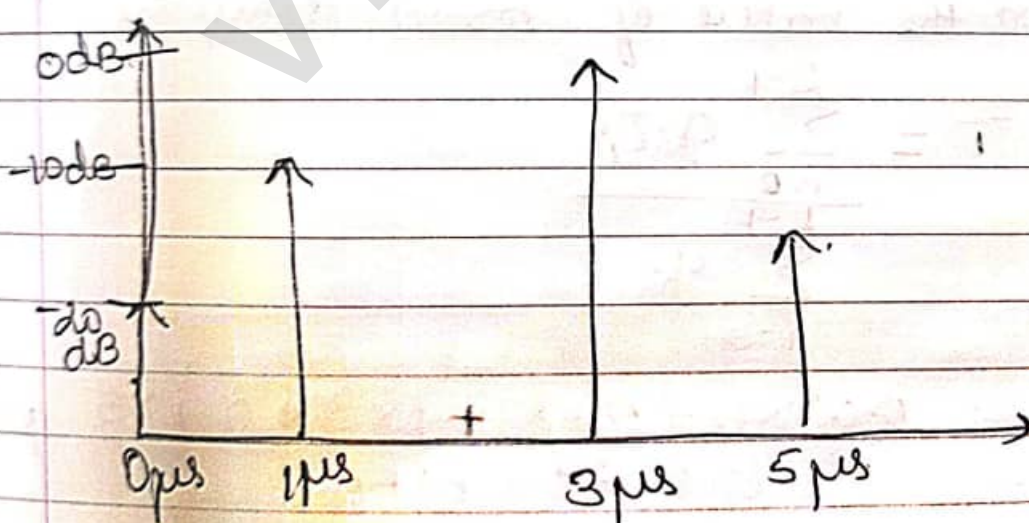
$$(\sigma_{\tau}^{\text{rms}})^2 = \sum_{i=0}^{L-1} b_i (\tau_i - \bar{\tau})^2$$

Substitute b_i

$$\sigma_T^{Rms} = \sqrt{\frac{\sum_{i=0}^{L-1} g_i (\tau_i - \bar{\tau})^2}{\sum_{i=0}^{L-1} g_i}}$$

$$= \sqrt{\frac{\sum_{i=0}^{L-1} |a_i|^2 (\tau_i - \bar{\tau})^2}{\sum_{i=0}^{L-1} |g_i|^2}}$$

prob) Consider the multipath power profile of a wireless channel shown in fig 4 comprising $L=4$ multipath components. Compute the Rms delay spread σ_T^{Rms} for this wireless channel.



1st path corresponding to $\tau_0 = 0\mu s$.

$$g(\text{dB}) = -20\text{dB}$$

$$10 \log_{10} g_0 = -20 \text{ dB}$$

$$\log_{10} g_0 = \frac{-20}{10} = -2$$

$$g_0 = 10^{-2} = 0.01$$

$$a_0 = \sqrt{g_0} = \sqrt{0.01} = 0.1$$

$$\therefore g_0 = 10a_0^2$$

$$a_0 = \sqrt{g_0}$$

W4 calculate for all values.

τ	dB gain	g	$a = \sqrt{g}$
0 μs	-20 dB	0.01	0.1
1 μs	-10 dB	0.1	0.3162
3 μs	0 dB	1	1
5 μs	-10 dB	0.1	0.3162

Find the value of mean delay.

$$\bar{T} = \frac{\sum_{i=0}^{L-1} g_i \tau_i}{\sum_{i=0}^{L-1} g_i}$$

$$= \frac{0.01 \times 0 + 0.1 \times 1 + 1 \times 3 + 0.1 \times 5}{0.01 + 0.1 + 1 + 0.1} \mu\text{s}$$

$$= \underline{\underline{2.9752 \mu\text{s}}}$$

Rms delay

$$\sigma_T^{\text{Rms}} = \sqrt{\frac{\sum_{i=0}^{L-1} g_i (\tau_i - \bar{\tau})^2}{\sum_{i=0}^{L-1} g_i}}$$

$$= \sqrt{\frac{0.01 \times (0 - 2.9752)^2 + 0.1 \times (1 - 2.9752)^2 + 1 \times (3 - 2.9752)^2 + 0.1 \times (5 - 2.9752)^2}{0.01 + 0.1 + 1 + 0.1}}$$

$$= 0.8573 \mu\text{s}$$

Rms delay is $0.8573 \mu\text{s}$ is more realistic than maximum delay spread
 $\sigma_T^{\text{max}} = 5 \mu\text{s}$.

Rms delay based on Average Power profile

Delay Spread metric to a continuous average power profile.

Consider $|h(\tau)|^2$ is instantaneous power corresponding to the delay τ .
 The average power associated with this delay

$$\phi(\tau) = E \{ |h(\tau)|^2 \}$$

$\phi(\tau)$ is average power associated with the

delay τ

Fractional power associated with delay τ is

$$f(\tau) = \frac{\phi(\tau)}{\int_0^{\infty} \phi(\tau) d\tau}$$

$f(\tau)$ is power distribution density corresponding to the delay.

$$\bar{\tau} = \int_0^{\infty} \tau f(\tau) d\tau$$

$$= \frac{\int_0^{\infty} \tau \phi(\tau) d\tau}{\int_0^{\infty} \phi(\tau) d\tau}$$

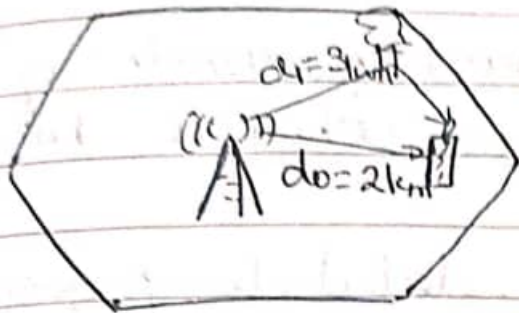
The RMS delay spread for the $\phi(\tau)$ profile

$$\sigma_{\tau}^{\text{RMS}} = \sqrt{\int_0^{\infty} (\tau - \bar{\tau})^2 f(\tau) d\tau}$$

$$= \sqrt{\frac{\int_0^{\infty} (\tau - \bar{\tau})^2 \phi(\tau) d\tau}{\int_0^{\infty} \phi(\tau) d\tau}}$$

4.2) Average delay Spread in outdoor cellular channels

Consider a outdoor cellular wireless communication system. The cell radius is in the range of 1-5 km



$$\tau_0 = \frac{2 \text{ km}}{c} \quad \tau_1 = \frac{3 \text{ km}}{c}$$

$$c = 3 \times 10^8 \text{ m/s.}$$

delay spread

$$\begin{aligned} \sigma_{\tau}^{\text{max}} &= \Delta z / c \\ &= \tau_1 - \tau_0 \\ &= \frac{\Delta d}{c} \end{aligned}$$

$$= \frac{3000 - 2000 \text{ m}}{3 \times 10^8}$$

$$= 3.33 \mu\text{s}$$

This calculation is not exactly calculation. But with this data ^{max} delay spread is in the range of $1 \mu\text{s} - 3 \mu\text{s}$.

In general, indoor delay spread is in the order of 10-50 ns.

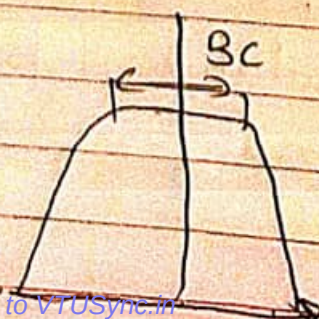
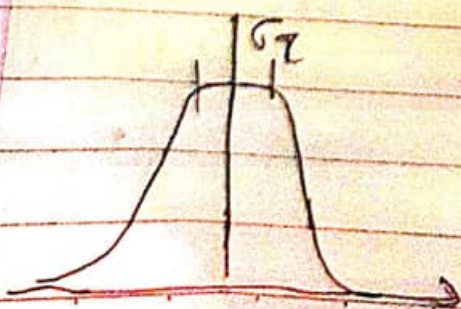
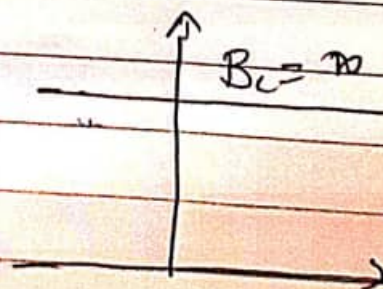
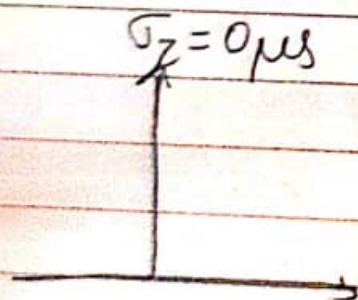
4.3 Coherence Bandwidth in Wireless Communications

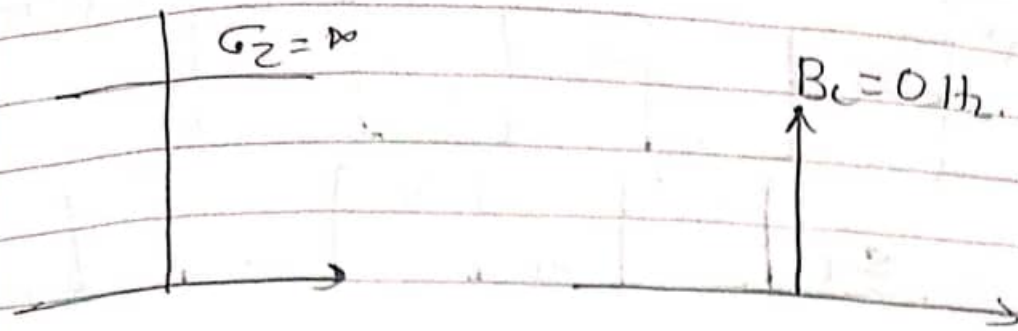
Another important parameter of wireless communication is, coherence bandwidth B_c .

Bandwidth is related to frequency. So, let us consider frequency response is defined by $H(f)$.

$$H(f) = \int_{-\infty}^{\infty} h(\tau) e^{-j2\pi f\tau} d\tau$$

In this section, we correlate the relation b/w coherence bandwidth & the delay spread.





$$B_c \approx \frac{1}{\sigma_\tau}$$

It shows that Bandwidth ↓ as delay Spread σ_τ ↑.

Consider delay spread is equal to $2\mu s$.
Find the value of B_c .

$$B_c = \frac{1}{2\mu s} = 250 \text{ KHz}$$

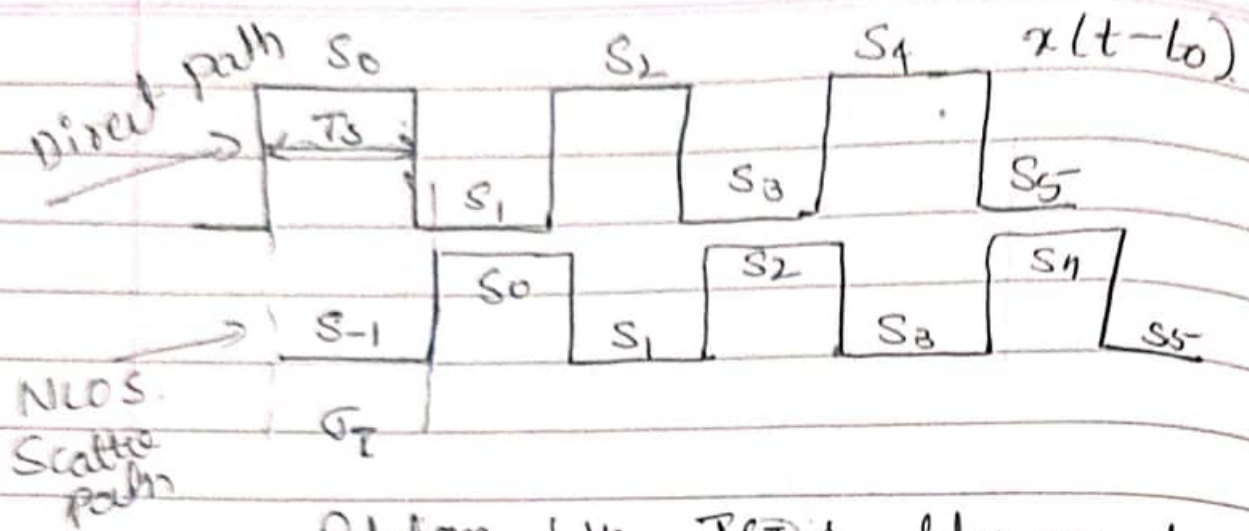
4.4 Relation b/w T_{SD} & Coherence B.W

ISI $\rightarrow$ Inter Symbol Interference

ISI & Coherence bandwidth are closely related concepts.

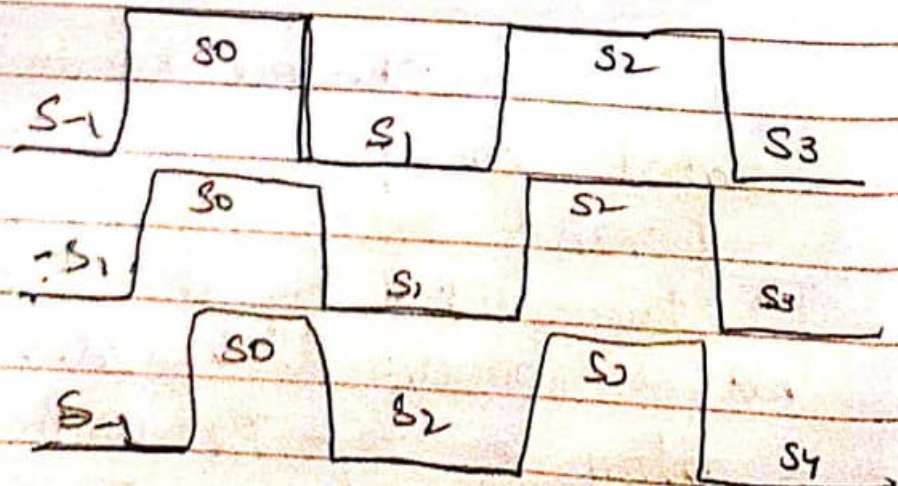
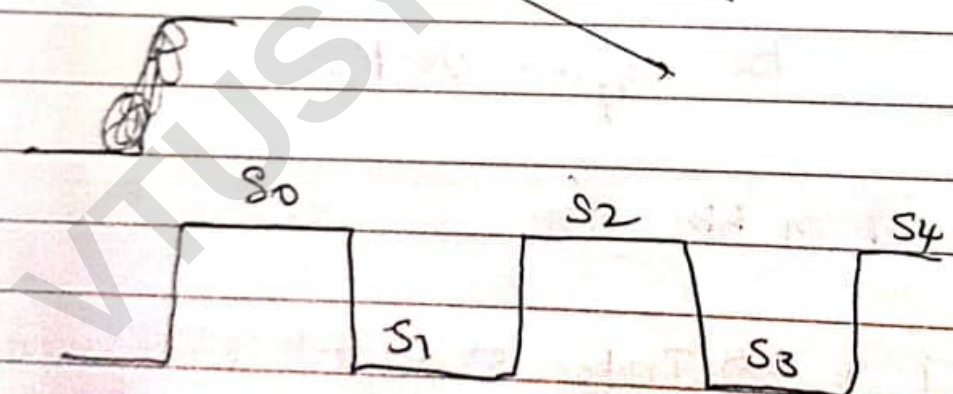
Consider PAM signal $x(t)$ of symbol T_s transmitted by the base station.

Let us consider the scatter component at a delay of $T_1 = T_d$ with addition to direct line of sight $T_0 = 0$.



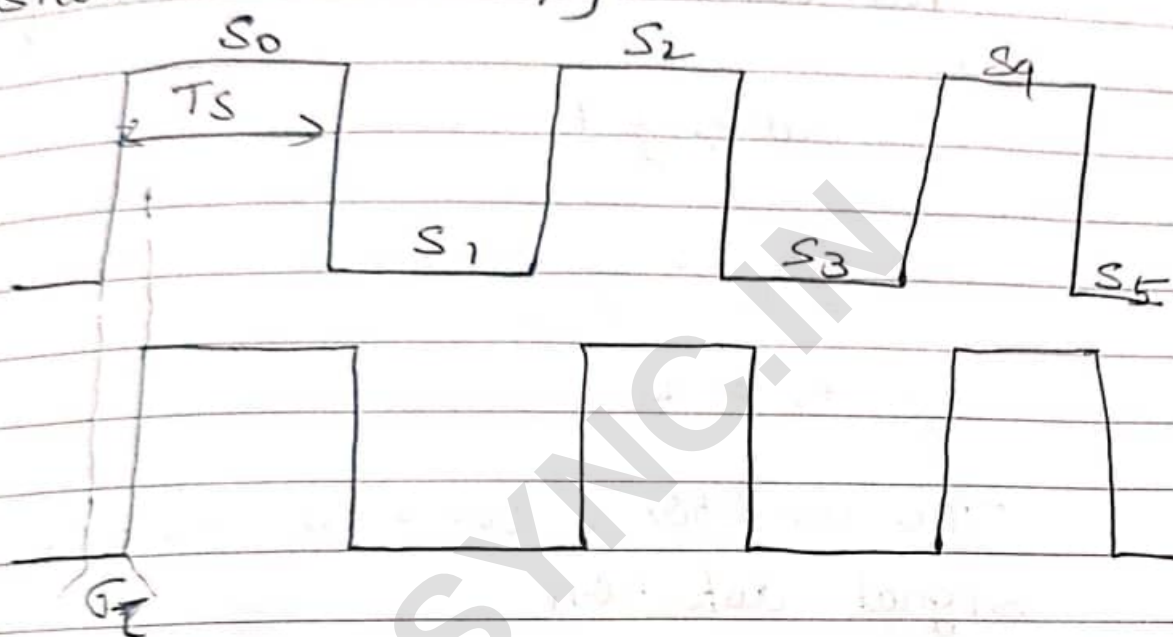
Relation btm ISI & delay spread

Way for multipath scatter component



ISI at the receiver is related to the interplay between the symbol time T_s & delay spread T_d .

For instance if symbol time T_s is much larger than delay spread T_d as shown in below fig



ISI negligible when $T_d \ll T_s$

~~The ISI will plays an i~~

The ISI occurs when delay spread T_d much greater than compared to T_s

$T_d \gg \frac{1}{2} T_s$ Consider the Scenario

The T_s is related bandwidth B_c as

$$T_s = \frac{1}{B_s}$$

The T_d is related to coherence BW as

$$B_c = \frac{1}{2} \frac{1}{T_d}$$

$$T_d = \frac{1}{2} \frac{1}{B_c}$$

The criteria for inter-symbol interference is

$$T_d > \frac{1}{2} T_s$$

$$\frac{1}{2} \frac{1}{B_c} > \frac{1}{2} \frac{1}{B_s}$$

$$\Rightarrow B_s \gg B_c$$

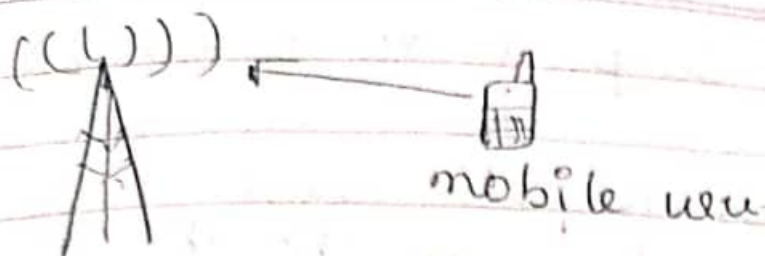
This condition is same as ^{frequency} Signal Selective distortion.

$\therefore$ frequency-selective distortion & ISI are ~~same~~ essentially both side of the same coin.

4.5 Doppler Fading in wireless systems

Doppler fading is small scale fading in wireless commⁿ. which is caused by relative motion bth the transmitter, & receive or surrounding object.

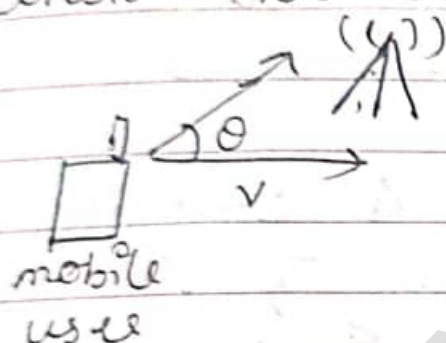
Doppler fading refers to signal amplitude fluctuation due to doppler effect. which occurs when either Tx or Rx are moving



Doppler fading due to user mobility

Doppler Shift computation

Consider the scenario as shown below



In this mobile station moving with velocity v at an angle θ with line of joining the mobile & base station. Let us consider carrier frequency f_c . The doppler shift is given by

$$f_d = \left(\frac{v}{c} \cos \theta \right) f_c$$

$c = 3 \times 10^8$ m/s velocity of light.
i.e. velocity of electromagnetic wave in free space

Doppler shift $\uparrow$ as with $\uparrow$ in velocity & also depends upon angle θ between the direction of motion & line joining transmitter & receiver.

4.6 Doppler Impact on a wireless channel

Consider impulse response of the i^{th} component of multipath channel is,

$$a_i \delta(t - \tau_i).$$

Now consider vehicle is moving with velocity v at an angle θ with respect to line joining the mobile & base station.

Since, mobile is changing constant due to the motion of the user the delay of i^{th} signal component is also changing.

Now consider initial distance for the i^{th} signal component be d_i , the initial propagation delay is

$$\tau_i = \frac{d_i}{c}$$

After small interval of time t , this distance decreases by $vt \cos \theta$.

$\therefore$ the term $v \cos \theta$ is the component of the velocity in the direction of the base station.

Time delay of i^{th} component after a small interval of time, t is, the

$$\tau_i(t) = \frac{d_i - vt \cos \theta}{c}$$

$$= \frac{d_i}{c} - \frac{vt}{c} \cos \theta$$

$$= \tau_i - \frac{vt}{c} \cos \theta$$

consider the expression of flat fading wireless channel coefficients.

$$h = \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i}$$

Now by considering the velocity component into account to derive the channel coefficient h as follows.

$$h(t) = \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i(t)}$$

$$= \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \left[\tau_i - \frac{vt \cos \theta}{c} \right]}$$

$$= \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i} \cdot e^{j \frac{2\pi f_c v \cos \theta}{c} t}$$

$$= \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i} e^{j2\pi f_d t}$$

$$\therefore f_d = f_c \frac{v \cos \theta}{c}$$

The channel coefficient contains the term $e^{j\phi(t)}$ which is the time-varying phase of the wireless channel.

The rate of variation of the phase is given by the doppler frequency f_d .

In summary, the mobility of the user in a wireless communication system leads to a doppler shift, which in turn results in a time-varying wireless channel coefficient. The time varying nature of the wireless channel is also termed time-selectivity & time-varying wireless channel is termed as time-selective channel.

In some wireless channel, can be both time & frequency selective. Such channel is termed as doubly selective wireless channels.

4.7 Coherence time of the wireless channel

Likewise ~~Ban.~~ coherence B.W, ^{is related} ~~some~~ to frequency-selective channel.

MM coherence time interval T_c is used for time-varying channel.

Consider the i^{th} multipath component of the time-varying channel coefficient.

$$a_i(t) = a_i e^{-j2\pi f_c t} e^{j2\pi f_d t}$$

Find the value of i^{th} component corresponding to $t=0$ & $t = \frac{1}{4f_d}$.

$$\begin{aligned} t=0 \Rightarrow a_i(0) &= a_i e^{-j2\pi f_c t} e^{j2\pi f_d(0)} \\ &= a_i e^{-j2\pi f_c t} \end{aligned}$$

$$t = \frac{1}{4f_d} \Rightarrow a_i e^{-j2\pi f_c t} e^{j2\pi f_d \frac{1}{4f_d}}$$

$$= a_i e^{-j2\pi f_c t} \cdot e^{j\frac{\pi}{2}}$$

$$= a_i e^{-j2\pi f_c t} \left[\cos \frac{\pi}{2} + j \sin \frac{\pi}{2} \right]$$

$$= a_i e^{-j2\pi f_c t} [0 + j]$$

$$= j a_i e^{-j2\pi f_c t}$$

It shows that for the values of

$$t=0 \quad \& \quad t = \frac{1}{4f_d}$$

for the phase angle vary from 0 to $\frac{\pi}{2}$.

This time duration in which channel changes significantly due to the mobility of the user is known as coherence time T_c .

f_d depends on the angle of motion θ ,

Minimum coherence time is obtained by setting $\theta = \frac{\pi}{2}$.

$$T_{c,min} = \frac{1}{4f_{d,max}}$$

w.k.T $f_d = \frac{v}{c} \cos \theta f_c$

$$f_{d,max} = \frac{v}{c} \cos\left(\frac{\pi}{2}\right)$$

$f_{d,max}$ will occur when $\cos \theta = 1$

$$f_{d,max} = \frac{v}{c} f_c$$

$$T_{c,min} = \frac{1}{4f_d}$$

Doppler Spread B_d ,

The doppler shift for single path is equal to the frequency shift caused by relative velocity.

In multipath environment, there are many shifts, so we call it as a doppler spectrum.

Doppler Spread B_d = range of frequencies shift due to motion.

$$\text{max doppler shift } f_d^{\text{max}} = \frac{v}{c} f_c$$

If doppler shifts range from

$$-f_d^{\text{max}} \text{ to } +f_d^{\text{max}}$$

$$B_d = 2f_d^{\text{max}}$$

So, the coherence time can be related to doppler spread as

$$T_c = \frac{1}{2B_d}$$

if we substitute $B_d = 2f_d^{\text{max}}$

$$T_c = \frac{1}{2(2f_d^{\text{max}})} = \frac{1}{4f_d^{\text{max}}}$$

Pb) consider a vehicle moving at 60 miles per hour at an angle of $\theta = 30^\circ$ with the line joining the base station. Compute the Doppler Shift of the received signal at a carrier frequency of $f_c = 1850 \text{ MHz}$.

Step 1: Convert 60 mph to m/s.

$$60 \text{ mph} = 60 \times 1.611$$

conversion of mph to m/s

$$1 \text{ mile} = 1609.344 \text{ meters}$$

$$1 \text{ hour} = 3600 \text{ Sec.}$$

$$\begin{aligned} x \text{ mph} &= x \times \frac{1609.34 \text{ m}}{1 \text{ mile} \times 3600} \text{ m/s} \\ &= \frac{60 \times 1609.34}{3600} \text{ m/s} \end{aligned}$$

Or easily we can remember that $1 \text{ mph} = 0.44704 \text{ m/s}$.

$$\begin{aligned} 60 \text{ mph} &= 60 \times 0.44704 \text{ m/s} \\ &= 26.82 \text{ m/s} \end{aligned}$$

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$$f_d = \left(\frac{v}{c} \cos \theta \right) f_c$$

$$= \frac{26.8}{3 \times 10^8} \times \cos(30^\circ) \times 1850 \times 10^6$$

$$= \underline{\underline{143 \text{ Hz}}}$$

Since the value is +ve indicating that mobile user moving towards the base station. So perceived frequency f_r is higher compared to the carrier frequency f_c . & is given by

$$f_r = f_c + f_d$$

$$= 1850 \text{ MHz} + 143 \text{ kHz}$$

(pb) For above problem find the coherence time T_c .

Soln $T_c = \frac{1}{4 f_d^{\max}}$

$f_d^{\max}$ is obtained when $\theta = 0$.

$$f_d^{\max} = \frac{v}{c} f_c$$

$$= \frac{26.8}{3 \times 10^8} \times 1850 \times 10^6$$

$$B_d = 2$$

$$T_c = \frac{1}{A \times 165}$$
$$= \underline{\underline{1.5 \text{ ms}}}$$

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Introduction to CDMA

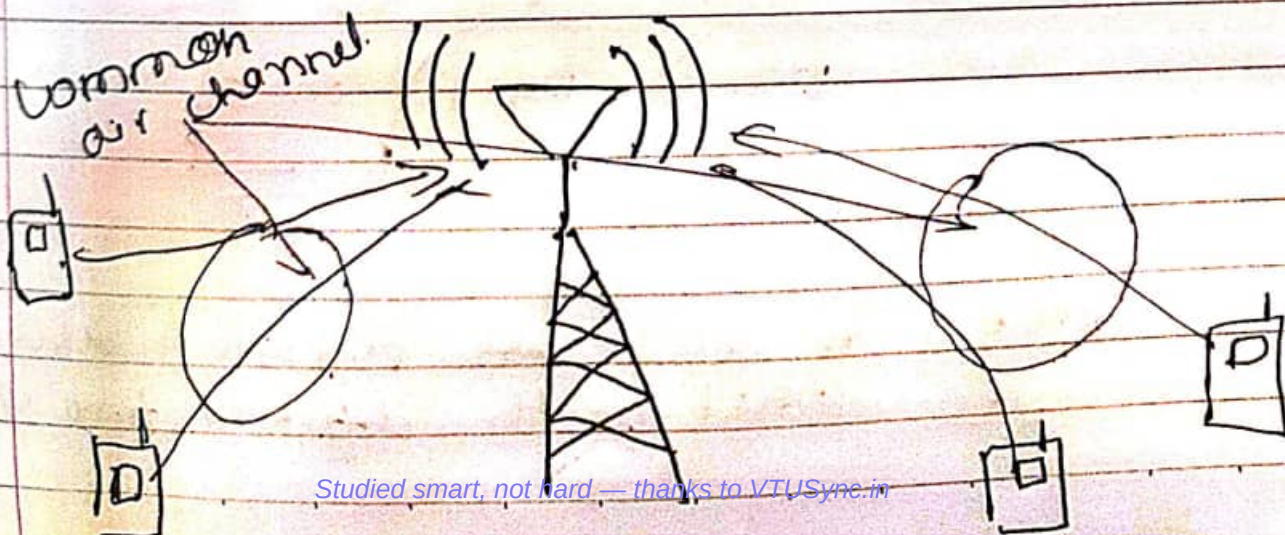
CDMA : code division ^{for} multiple Access.

→ It is a path-breaking wireless technology.

→ 1st introduced in the 2nd generation IS-95 cellular Standard. used in North America. Under brand name cdmaOne.

In conventional ~~wireless~~ wired communication system, there is a dedicated wireline channel which is used to connect particular device such as telephone to base station.

But in case of wireless commn, devices are required to share the common radio channel over the air. as shown in below figure.



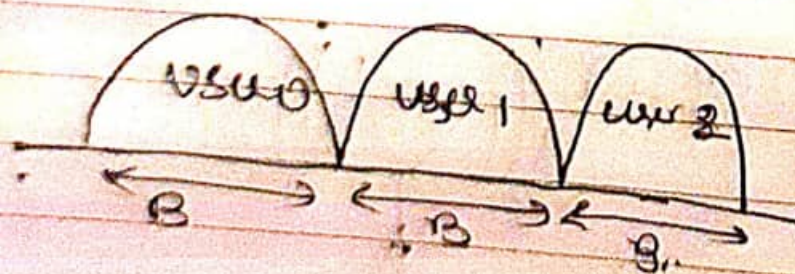
In this radio channel is common for all the user/devices & the available frequency band is limited.

So, it is necessary to devise a mechanism for multiple users to access this common radio channel which is termed as multiple access technology. Thus, multiple access is the heart of the modern wireless technology especially 3G & 4G cellular technologies.

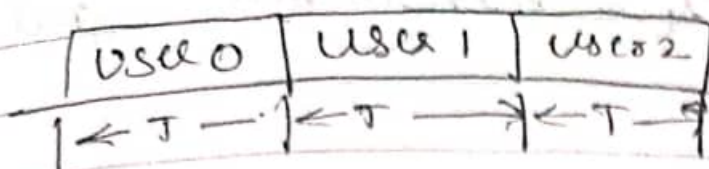
Several multiple access technologies are

- FDMA
- TDMA
- CDMA

FDMA → Different users are allotted different frequency band.



TDMA: In this use different users are allotted with different time slots.



Basic CDMA mechanism

CDMA is multiple access technology based on code division.

In this different users are allocated with ~~the~~ different codes.

Consider two-user scenario.

i.e. two users accessing the radio channel simultaneously.

Let a_0 is the symbol used for user 0 & a_1 is used for user 1.

Consider code $C_0 = [1, 1, 1, 1]$ is unique code used for user 0.

$C_1 = [1, -1, -1, 1]$ is used for user 1.

~~Eq~~

Now let us consider x_0 is the transmitted signal of the user 0 obtained by multiplying symbol a_0 & unique code C_0 .

$$x_0 = a_0 C_0$$

$$= a_0 [1, 1, 1, 1]$$

$$= [a_0, a_0, a_0, a_0]$$

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Now x_1 is transmitted signal for user 1

$$\begin{aligned}x_1 &= a_1 c_1 \\&= a_1 [1, -1, -1, 1] \\&= [a_1, -a_1, -a_1, a_1]\end{aligned}$$

Both signals x_0 & x_1 are transmitted over a same channel as.

$$\begin{aligned}x &= x_0 + x_1 \\&= [(a_0 + a_1), (a_0 - a_1), (a_0 - a_1), (a_0 + a_1)]\end{aligned}$$

At the receiver end same unique code is used for user 0 & user 1 as c_0 & for user 1 as c_1

The received signal x is multiplied with c_0 .

$$\begin{array}{cccc} & a_0 + a_1 & a_0 - a_1 & a_0 - a_1 & a_0 + a_1 \\ \times & 1 & 1 & 1 & 1 \end{array}$$

$$(a_0 + a_1) + (a_0 - a_1) + (a_0 - a_1) + (a_0 + a_1)$$

$$= 4a_0$$

It contains a_0 Symbol which means User 0 receive the signal.

U4 for use 1.

$$\begin{array}{cccc} a_0 + a_1 & a_0 - a_1 & a_0 - a_1 & a_0 + a_1 \\ \times & 1 & -1 & -1 & 01 \\ \hline a_0 + a_1 & -a_0 + a_1 & -a_0 + a_1 & a_0 + a_1 \\ & = & 4a_1 & \end{array}$$

only User 1 received the signal.

∴ The key features of CDMA can be summarized as

- 1) multiplying or modulation the symbols of the different users with the corresponding assigned unique code.
- 2) combining or adding the code-modulated signals of all the users to form the composite signal.
- 3) Finally, correlation of the composite received signal x at each user with the corresponding code of the user to recover the transmitted symbol.

Fundamentals of CDMA codes

Now compute the correlation γ_{01} of user code C_0 & yields

$$\gamma_{01} = \sum_{k=0}^3 C_0(k) C_1(k)$$

$$= (1 \times 1) + (1 \times -1) + (-1 \times 1) + (1 \times 1)$$

$$= 1 - 1 - 1 + 1$$

$$= 0$$

Since, the correlation between C_0 & C_1 is zero, the codes are orthogonal. This property helps to recover the symbols of the different users from the composite signal.

This property is the key property of the codes employed in CDMA wireless channel system.

Let us consider the symbol rate for the symbols A_0 of the user 0 is 1 kbps.

So, time period per symbol is

$$T = \frac{1}{1 \text{ kbps}} = 1 \text{ ms}$$

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Corresponding B.W is,

$$B = \frac{1}{T}$$
$$= \frac{1}{1 \text{ ms}} = 1 \text{ kHz}$$

During the transmission of the symbol a_0 is multiplied with corresponding code C_0 ,

$$\text{i.e. } a_0 [1 \ 1 \ 1 \ 1] = [a_0 \ a_0 \ a_0 \ a_0]$$

So, each symbol a_0 , one has to transmit 4 chips. To keep the symbol rate constant at 1 kbps, the time of each chip T_c has to be set as $T_c = \frac{1 \cdot T}{4}$.

$$T_c = \frac{1}{4} (1 \text{ ms})$$

$$= 0.25 \text{ ms}$$

Thus the B.W required for this system is

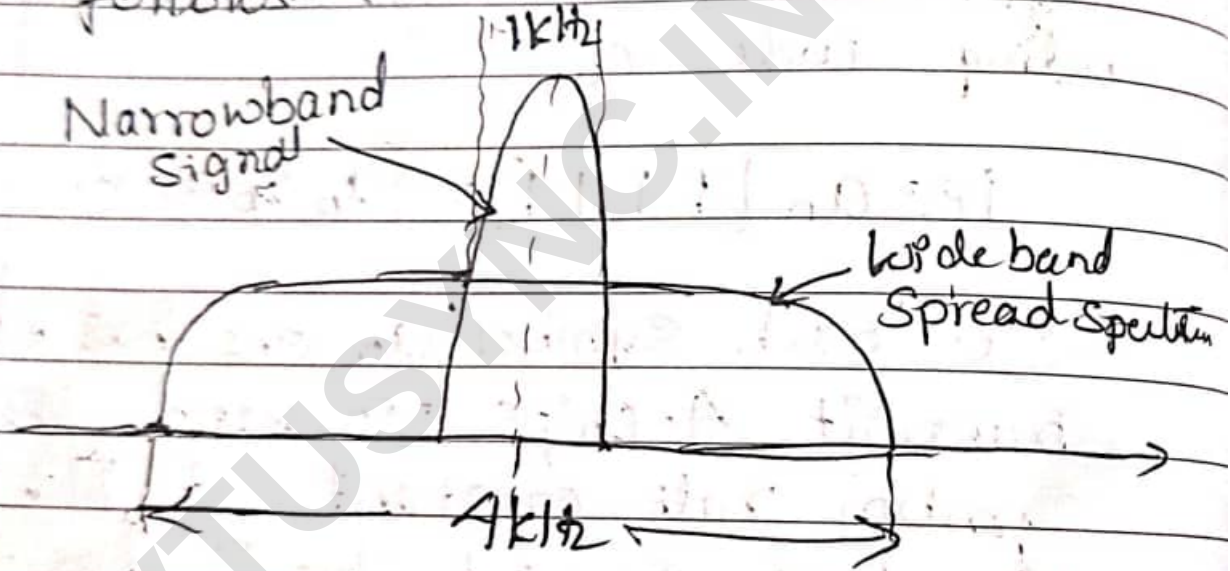
$$B_{\text{DMA}} = \frac{1}{T_c}$$

$$= \frac{1}{0.25 \text{ ms}} = 4 \text{ kHz}$$

Thus, modulating with code C of length $N=4$, results in an increase of the required B.W by a factor of N .

i.e from 1kHz to 4kHz

which can be Schematicly Shown as follows.



Spread spectrum communication

~~So~~ The original signal bandwidth results in spreading, hence it is termed as spreading code.

& also resulting signal occupies a large bandwidth, CDMA Systems are also called as Spread spectrum or wideband systems.

Consider the ~~leno~~ spreading code length
is N

$$N=4$$

So, the ϕ different orthogonal
spreading codes are

$$C_0 = 1 \ 1 \ 1 \ 1$$

$$C_1 = 1 \ -1 \ -1 \ 1$$

$$C_2 = 1 \ -1 \ 1 \ -1$$

$$C_3 = 1 \ 1 \ -1 \ -1$$

we can check the Orthogonal property
of code C_1 C_2 .

$$r_{12} = \sum_{k=0}^3 C_1(k) C_2(k)$$

$$= (1 \times 1) + (-1 \times -1) + (-1 \times 1) + (1 \times -1)$$

$$= 1 + 1 - 1 - 1$$

$$= 0$$

This implies that for a spreading
sequence length N , there exist N
orthogonal codes.

~~Even though~~ In this, the Bandwidth ~~res.~~
eventhough this technique is used

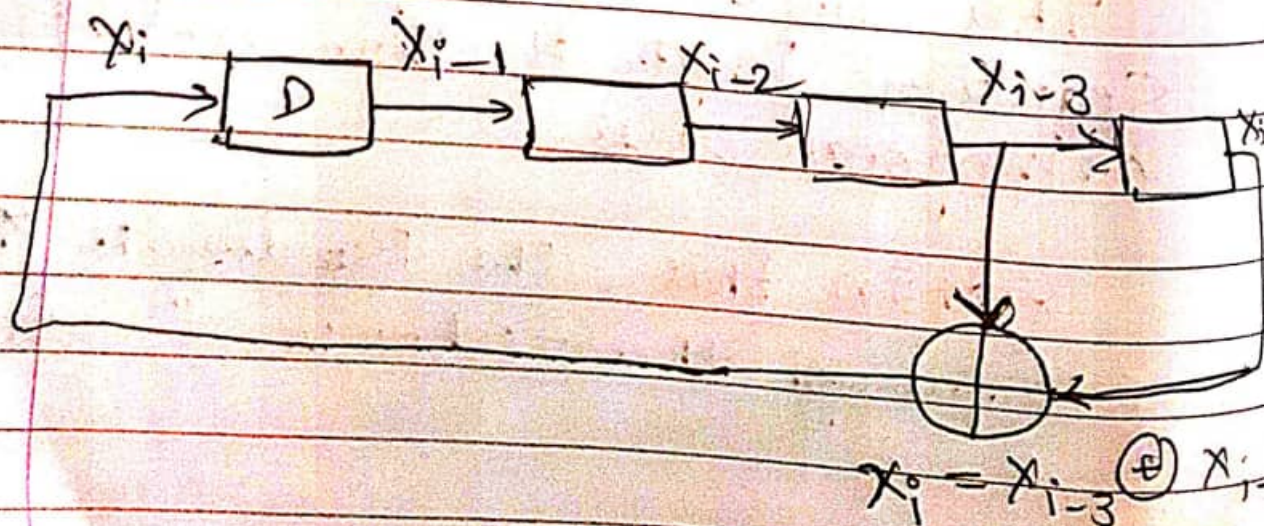
5.4 Spreading codes based on Pseudo-Noise [PN] Sequences

Consider the code $C_2 = [1, -1, 1, -1]$

In this code looks like a random sequence of $+1, -1$ or it is also known as pseudo-noise [PN] sequence because it resembles a noise sequence but is not actually a noise sequence.

A method which is used to generate a long spreading codes based on PN sequences is known as linear Feedback shift register [LFSR].

Consider the shift register architecture as shown in figure



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In above figure D is the delay element, & X_i is fed back to X_{p-1}

$$X_p = X_{i-3} \oplus X_{i-4}$$

which is the linear equation & it is feedback & uses delay elements or shift registers.

So, such circuit is called as a linear feedback Shift Register. [LFSR] architecture.

The above LFSR can generate sequence 15 states.

1111, 0111, 0011, 0001, 1000, 0100, 0010, 1001, 1100, 0110, 1011, 0101, 1010, 1101, 1110,

States can be calculated as.

$$2^D - 1 = 2^4 - 1$$

$$= 16 - 1 = 15 \text{ states}$$

The maximum state is $2^D = 2^4 = 16$

but if we take state as 0000, every time LFSR generates 0000, & will ~~not~~ never get out of the all zero's state. i.e. LFSR is undesirable. LFSR should not enter all zero state.

Studied smart, not hard - thanks to VTUSW.in

For BPSK, consider the o/p
 X_{i-4} is given by

PN sequence = 111100010011010

We can assume bits 1, 0 to the
BPSK symbols $-1, +1$ then modulated
PN sequence becomes

PN sequence = $-1 -1 -1 -1 +1 +1 +1 -1 +1 +1$
 $-1 -1 +1 -1 +1$

Properties of PN sequence

Property 1: Balance property

Consider the BPSK modulated
PN sequence, count -1 , & $+1$

There is 8 -1 ones & 7 $+1$ ones

It shows that number of -1 s is
one more than the number of $+1$ s.
This is known as balance property
of the PN sequence.

If we are generating random
noise of $+1, -1$ chips with

$$P(X_i = +1) = P(X_i = -1) = \frac{1}{2}$$

Then, ~~an~~ average that half the chips are +1 & the rest are -1.

In the above scenario, the number of chips is odd no that is 15, so it is not possible to get exact half chips are +1, so the value of +1 are 7 & remaining chips are -1 [i.e. 8] which is nearly equal to 7.

Property 2: Run-length Property

The run is a string of continuous values.

consider the sequence

$-1, -1, -1, -1, +1, +1, +1, -1, +1, +1,$
① $-1, -1, +1, -1, +1,$

1st run : $-1, +1, -1, -1 \rightarrow$ one
length 4

2nd run : $+1, +1, +1 \rightarrow$ one
length 3

3rd run : $-1, -1$ & $+1, +1 \rightarrow$ two

4th run : $-1, -1$ } -1 repeated two
 $+1, +1$ } +1 repeated two
total 8

It can also be found by using the formula

$$2^{(4-1)} = 2^{(4-1)} \\ = 8$$

Property 3

Correlation Property

Consider the BPSK Sequence.

-1 -1 -1 -1 +1 +1 +1 -1 +1 +1 -1 -1 +1 -1 +1
& it is denoted by $c_0(n)$

The Correlation is given as.

$$r_{00}(0) = \frac{1}{N} \sum_{n=0}^{N-1} c_0(n) c_0(n)$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} 1$$

$$= \frac{1}{N} [N] = 1$$

Now consider Shifted Version of $c_0(n)$ by 2 $c_0(n-2)$ where 2 indicates that the sequence is shifted by 2.

Shift the BPSK sequence by 2.
as follows.

~~1st shift~~:

Original sequence: -1 -1 -1 -1 +1 +1 +1 -1 +1 +1 -1 -1 +1 -1 +1

1st shift

Shifted sequence: +1 -1 -1 -1 -1 +1 +1 +1 -1 +1 +1 -1 -1 +1 -1

2nd shift

Shifted sequence: -1 + -1 -1 -1 -1 +1 +1 +1 -1 +1 +1 -1 -1 +1

Now find the correlation. b/w $c(n)$ & $c(n-2)$ by $r_{00}(2)$.

$$r_{00}(2) = \frac{1}{N} \sum_{n=0}^{N-1} c(n) \cdot c(n-2)$$

$$= \frac{1}{N} [(-1)(-1) + (-1)(1) + (-1)(-1) + (-1)(-1) + (1)(-1) + (1)(-1) + (1)(1) + (1)(1) + (-1)(1) + (-1)(-1) + (-1)(-1) + (-1)(-1) + (1)(1) + (1)(-1) + (1)(-1) + (1)(1)]$$

$$= \frac{1}{N} [(-1)(-1) + (-1)(1) + (-1)(-1) + (-1)(-1) + (1)(-1) + (1)(-1) + (1)(1) + (1)(1) + (-1)(1) + (-1)(-1) + (-1)(-1) + (-1)(-1) + (1)(1) + (1)(-1) + (1)(-1) + (1)(1)]$$

$$= \frac{1}{15} [1 - 1 + 1 + 1 - 1 - 1 + 1 - 1 + 1 - 1 - 1 - 1 - 1 + 1 + 1]$$

$$= -1/15$$

In general, $-\frac{1}{N}$.

∴ The autocorrelation property of the PN sequence is

$$r_{00}(n_0) = \frac{1}{N} \sum_{i=0}^{N-1} = \begin{cases} 1 & \text{if } n_0=0 \\ -\frac{1}{N} & \text{otherwise} \end{cases}$$

5.5 ~~Correlation~~

Correlation properties of Random CDMA Spreading Sequences

Consider the chip sequence $c_k(i)$, $0 \leq i \leq N-1$ with a probability as follows:

$$P(c_k(i) = +1) = P(c_k(i) = -1) = \frac{1}{2}$$

$$E\{c_k(i)\} = \frac{1}{2}(+1) + \frac{1}{2}(-1) = 0$$

Another important aspect is to choose a sequence that contains Independent Identically distributed ~~bits~~ $[+1, -1]$ chips.

that satisfy the condition or property

$$\begin{aligned} E \{c_k(i)c_k(j)\} &= E \{c_k(i)\} E \{c_k(j)\} \\ &= 0 \times 0 \\ &= \underline{\underline{0}} \end{aligned}$$

The above property implies that each chip $c_k(i)$ is uncorrelated with chip $c_k(j)$.

Consider independent sequence for different users.

$$E \{c_k(i)c_l(j)\} = E \{c_k(i)\} E \{c_l(j)\}$$

Now examine the correlation properties of random spreading sequence.

$r_{00}(k)$ denotes the autocorrelation properties of the chip sequence. $k=0$,

$$r_{00}(k) = \frac{1}{N} \sum_{i=0}^{N-1} c_0(i)c_0(i-k)$$

The average or expected value of $r_{00}(k)$ is given by

$$E\{r_{00}(k)\} = E\left\{\frac{1}{N} \sum_{i=0}^{N-1} c_0(i) c_0(i-k)\right\}$$

$$= \frac{1}{N} \sum_{i=0}^{N-1} E\{c_0(i) c_0(i-k)\}$$

$$= \frac{1}{N} \sum_{i=0}^{N-1} E\{c_0(i)\} E\{c_0(i-k)\}$$

$$= \frac{1}{N} \sum_{i=0}^{N-1} 0 \times 0$$

$$= 0$$

The average value or expected value of the correlation $E\{r_{00}(k)\}$ is zero for lag $k \neq 0$. This satisfies the random properties of spreading sequence.